CertificateAuthority

System Description

Abstract
This document describes the CertificateAuthority support system of the Eclipse Arrowhead 5.2.2 framework, as implemented in `ah5-CertificateAuthority-java-spring` with an independent local-cloud deployment definition in its own folder.

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1 Overview

This document describes the CertificateAuthority system. The CertificateAuthority is the local-cloud CA: it signs certificates under the Arrowhead profiles (onboarding, device, system, operator), validates certificates and trusted keys, keeps the certificate register and publishes the CRL.



Figure 1: SysML-style block view of CertificateAuthority with its produced and consumed services.

The rest of this document is organized as follows. In Section 1.1, we reference major prior art capabilities of the system. In Section 1.2, we describe the intended usage of the system. In Section 1.3, we describe fundamental properties provided by the system. In Section 1.4, we describe delimitations of capabilities of the system. In Section 2, we describe the abstract service functions consumed or produced by the system. In Section 3, we describe the security capabilities of the system.

1.1 Significant Prior Art

Successor of the Arrowhead 4.6 Certificate Authority; native Java port of the former Python implementation (the Python sources in this folder are kept only for rollback).

1.2 How This System Is Meant to Be Used

Onboarding requests certificates for devices and systems; Sysop manages the register and revocations; brokers, relay and systems use the CRL.

1.3 System functionalities and properties

1.3.1 Functional properties of the system

- CSR signing with proof of possession; subject and extensions built by the CA.
- Profiles per caller; reserved names; validity limits.
- Certificate and trusted-key validation.
- Register, revocation (also by content) and signed CRL publication.

1.3.2 Configuration of system properties

The system is configured with environment variables. The variables that shape its behaviour are:

Property	Default	Meaning
`PORT`	8448	HTTPS port (mutual TLS).
`TLS_CA_CERT` / `TLS_CA_KEY`	/certs/ca.*	Cloud CA certificate and key.
`TLS_CERT` / `TLS_KEY`	(required)	CertificateAuthority system certificate.
`CERT_VALIDITY_DAYS` / `ONBOARDING_CERT_VALIDITY_HOURS`	365 / 24	Validity limits.
`CRL_DIR` / `CRL_VALIDITY_HOURS`	(unset) / 168	CRL publication.
`RESERVED_NAMES`	core and support systems	Names only Sysop may request.
`SERVICE_REGISTRY_URL`	`https://serviceregistry:8443`	Registry for registration and identity lookup.
`AMQP_HOST` / `AMQP_PORT`	amqp-broker / 5671	sysMonitor publication.

Secrets (database, keystore and identity passwords, secret keys) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets under `/run/secrets`.

1.3.3 Data stored by the system

Issued certificates and trusted keys (SQLite).

1.3.4 Non functional properties

Every protected request verifies the caller's identity token with Authentication.

1.3.5 Stateful or stateless

Stateful.

1.4 Important Delimitations

No automatic renewal; no OCSP. The CA key is read from a file mount.

2 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

2.1 Produced services

Service	Version	Interfaces	SD	IDD
`certificateSigning`	1.0.0	HTTP(S)	[1]	[2]
`certificateValidation`	1.0.0	HTTP(S)	[3]	[4]
`certificateAuthority Management`	1.0.0	HTTP(S)	[5]	[6]
`sysMonitor`	1.0.0	AMQP(S)	[7]	[8]

All services use `generic_https` with `CERT_AUTH`; `sysMonitor` uses `generic_amqps`. The container also answers `GET /health` on a loopback-only plain-HTTP port for the container health check.

2.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	systemDiscovery, serviceDiscovery	register, lookup	Registration; lookup of Authentication.	always
Authentication	identity	identity-login, identity-verify	Own token; caller verification.	always
AMQP broker (SysMonitor)	sysMonitor (publisher)	echo, power, status	Own observations.	always

Only Authentication's `identity` and the orchestrator are looked up in the ServiceRegistry; other services are reached through DynamicServiceOrchestration (results cached), and every such call needs a ConsumerAuthorization grant.

3 Security

The standalone Java system requires HTTPS with mutual TLS and X.509 certificates following the Arrowhead system certificate profile (`<System>.<Cloud>.<Organization>.arrowhead.eu`, DNQ `sy`). There is no insecure runtime mode. Identity tokens are verified through Authentication and bound to the caller certificate. A separate loopback health endpoint is infrastructure only. Provision credentials and a valid CRL using the project-local environment template and Compose definition.

3.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mandatory client certificates; MQTT over TLS; AMQP over TLS for `sysMonitor`.
- **Data protection:** transport encryption by TLS; SQLite storage is local to this system's private volume. Passwords are read from Compose file secrets; certificate and private-key files are mounted individually.
- **System authentication:** mutual TLS with a system or operator certificate of the cloud and an identity token of the same system, verified with Authentication; declared identities are not accepted. Client certificates are checked against the CRL.
- **Authorization of produced services:** per operation by caller lists in the implementation and, where stated, by ConsumerAuthorization.
- Callers need a system or operator certificate and an identity token of the same system. Signing only for Onboarding, Sysop and CertificateAuthority; management only Sysop and CertificateAuthority.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

4 References

[1] J. Delsing, "certificateSigning – Service Description (CertificateAuthority)," Eclipse Arrowhead, SD 5.2.2, 2026, certificateAuthority-v5.2.2-certificateSigning-SD.pdf.

[2] ——, "certificateSigning HTTP/TLS/JSON – Interface Design Description (CertificateAuthority)," Eclipse Arrowhead, IDD 5.2.2, 2026, certificateAuthority-v5.2.2-certificateSigning-HTTP-IDD.pdf.

[3] ——, "certificateValidation – Service Description (CertificateAuthority)," Eclipse Arrowhead, SD 5.2.2, 2026, certificateAuthority-v5.2.2-certificateValidation-SD.pdf.

[4] ——, "certificateValidation HTTP/TLS/JSON – Interface Design Description (CertificateAuthority)," Eclipse Arrowhead, IDD 5.2.2, 2026, certificateAuthority-v5.2.2-certificateValidation-HTTP-IDD.pdf.

[5] ——, "certificateAuthorityManagement – Service Description (CertificateAuthority)," Eclipse Arrowhead, SD 5.2.2, 2026, certificateAuthority-v5.2.2-certificateAuthorityManagement-SD.pdf.

[6] ——, "certificateAuthorityManagement HTTP/TLS/JSON – Interface Design Description (CertificateAuthority)," Eclipse Arrowhead, IDD 5.2.2, 2026, certificateAuthority-v5.2.2-certificateAuthorityManagement-HTTP-IDD.pdf.

[7] ——, "sysMonitor – Service Description (CertificateAuthority)," Eclipse Arrowhead, SD 5.2.2, 2026, certificateAuthority-v5.2.2-sysMonitor-SD.pdf.

[8] ——, "sysMonitor AMQP/TLS/JSON – Interface Design Description (CertificateAuthority)," Eclipse Arrowhead, IDD 5.2.2, 2026, certificateAuthority-v5.2.2-sysMonitor-AMQP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-25	5.2.2	Independent Java build, private helpers and state, per-system deployment and configuration.	Codex (draft)
3	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	